

Article

Serratia marcescens Is Associated with Larval Mortality and Gut Dysbiosis in the Hornet *Vespa analis*

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Simple Summary

Hornets are usually known as dangerous or invasive insects, but in some parts of China, they are also reared for food and traditional medicine and can provide important income for local farmers. In indoor farms in Dehong, Yunnan, outbreaks of disease in larvae of the hornet *Vespa analis* have caused yellow bodies, weakness, larvae falling from the nests, and high mortality, yet the cause was unknown. In this study, we used microbiological methods and feeding experiments to identify a candidate bacterial factor associated with outbreaks. We isolated a red bacterium from the guts of sick larvae, identified it as *Serratia marcescens*, and showed that feeding this bacterium to healthy colonies produced similar signs and deaths. We also found that the normal gut microbes became disturbed during infection. These results show that *S. marcescens* is a key factor in larval disease in *V. analis* and provide a scientific basis for better hygiene, pathogen monitoring, and health management in hornet rearing.

Abstract

Social wasps, including hornets, are increasingly recognized not only as invasive pests but also as farmed insects; however, their gut microbiota and associated diseases remain poorly characterized. In indoor rearing facilities for the hornet *Vespa analis* in Dehong, Yunnan, China, we observed recurrent larval disease with weakness, larvae falling from the nests, and high mortality. To identify the causative agent and its effects on the gut community, we isolated bacteria from diseased larvae, characterized them by morphology, biochemical tests, and 16S rRNA gene sequencing, and then established an oral infection model. A red-pigmented isolate, designated YR2, was identified as *Serratia marcescens*. Oral inoculation with YR2 reproduced disease signs and significantly increased larval mortality, and a phenotypically consistent *S. marcescens* isolate was reisolated from infected larval guts. Amplicon sequencing showed that healthy larvae harbored gut communities dominated by Proteobacteria, whereas infection was associated with reduced diversity and a dysbiotic shift with enrichment of Enterobacterales. Our results support *S. marcescens* as a strong candidate pathogen associated with larval disease and mortality in *Vespa analis* under indoor-rearing conditions. Our findings provide a basis for pathogen surveillance and microbiota management in indoor hornet husbandry, and support improved biosecurity and health monitoring practices.



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Keywords: *Vespa analis*; hornet husbandry; *Serratia marcescens*; insect gut microbiota; hornet industry

1. Introduction

Social wasps (Hymenoptera: Vespidae) are widespread and ecologically prominent, functioning both as key terrestrial predators and as occasional pollinators, thereby helping to stabilize food webs and promote plant reproduction [1,2]. Nevertheless, the coexistence of ecosystem services and disservices complicates human–wasp interactions [2]. Notably, species within the genera *Vespa*, *Vespula*, and *Polistes* are recognized as globally important invasive pests that threaten honey bee populations and the apiculture industry, while also posing substantial risks to biodiversity and human health [3,4]. For example, the Asian hornet *Vespa velutina* has spread rapidly across Europe and Asia, causing large-scale losses of honey bees and exerting persistent pressure on local ecosystems [3]. Conventional control relies on chemical insecticides. Such approaches lack specificity, cause environmental contamination and nontarget impacts, and accelerate resistance evolution when used persistently [5,6]. Consequently, there is a pressing need for safe, sustainable, and environmentally friendly management strategies.

Beyond potential ecological risks, artificial rearing of social wasps has expanded in several regions, for example, in Dehong, Yunnan, China, providing edible and medicinal products that yield substantial income for local communities; however, based on our observations in Dehong, recurrent disease during husbandry has become a critical bottleneck that restricts industry development. In indoor settings, the typical clinical course includes larval yellowing, reduced feeding, weakness followed by dropping from the comb, slowed foraging by workers, anorexia and diarrhea in queens and workers, and a sustained decline in colony vigor; transmission across colonies within the same room can occur in approximately 7 days, precipitating outbreaks that involve the entire room and cause significant economic losses. To elucidate the etiology and propose actionable control measures, we conducted gut microbiological screening and pathogenicity verification in colonies that exhibited these representative signs.

In recent years, insect-associated microbiomes have become a major focus in the life sciences, and abundant evidence shows that the gut microbiota of insects plays central roles in host nutrition and metabolism, immune regulation, pathogen defense, and development [7]. At the same time, the insect gut can serve as a cryptic reservoir for pathogens, and under specific conditions, certain bacteria may cross host barriers or deploy virulence factors, leading to bacteremia or septicemia and ultimately mortality [8]. This understanding not only deepens insights into insect health but also provides new avenues for microbe-based biological control of pests [8]. In pollinators such as the honey bee *Apis mellifera* and bumblebees *Bombus* spp., potential pathogens including *Pseudomonas*, *Serratia*, *Enterococcus*, and *Bacillus* have been reported to damage gut tissues or induce dysbiosis, thereby causing disease at individual and even colony scales [9]. Particularly, *S. marcescens* is a ubiquitous Gram-negative bacterium widely distributed in soil, water, and insect-associated environments [10]. In insects, it is frequently reported as an opportunistic or context-dependent pathogen, with documented pathogenicity in several Hymenopteran hosts, including honey bees and hornets, although virulence often varies among strains and experimental conditions [11,12].

However, compared with model or economically important insects like bees, research on the gut microbiota of social wasps remains nascent. Prior studies have largely focused on descriptive assessments of community composition and diversity, relying mainly on

16S rRNA operational taxonomic units or amplicon sequence variants without direct experimental evidence for pathogenic potential [12–14]. Consequently, a robust causality chain for gut pathogens in social wasps (including hornets) is still lacking.

Accordingly, we used larvae of the hornet *Vespa analis* as the study system, isolated and identified candidate pathogens from the guts of naturally diseased individuals; established a colony-level oral infection model to assess clinical signs and survival under standardized doses; and reisolated a phenotypically consistent strain from infected larvae, completing key experimental steps consistent with Koch’s framework (isolation–oral challenge–reisolation), while acknowledging that strain-level genetic confirmation (e.g., MLST or whole-genome comparison) remains to be completed. In parallel, we performed 16S rRNA amplicon sequencing to determine whether infection induced dysbiosis and to characterize its phylogenetic signature. Integrating oral challenge assays with gut microbiome profiling, this study provides an evidence-based foundation for pathogen surveillance and microbiota-informed health management in indoor hornet husbandry.

2. Materials and Methods

2.1. Source and Husbandry of Hornets

This study was conducted on indoor-reared colonies of the hornet *V. analis*. In Dehong Prefecture, Yunnan, hornet rearing is predominantly performed indoors, with approximately 1000 colonies maintained per rearing room. At the early stage of disease outbreaks, 3–6 colonies per room typically exhibit characteristic clinical signs; if timely control measures are not implemented, most colonies in the same room (up to approximately 90%) may subsequently become affected within about one week. The indoor-reared *V. analis* nests and colony-level housing units is shown in Figure 1.

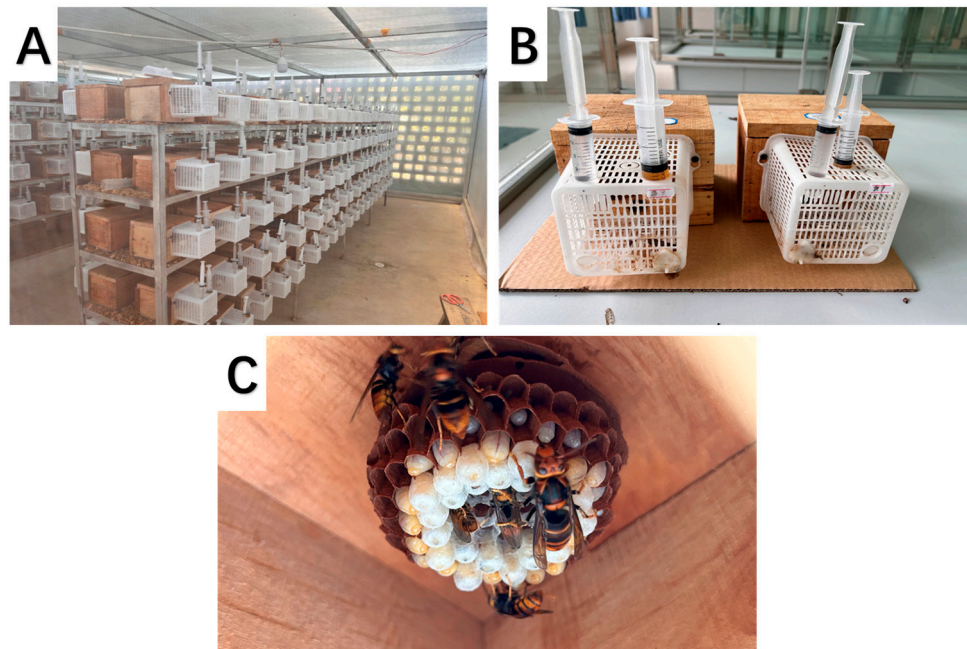


Figure 1. Indoor rearing conditions and colony housing of *V. analis*. (A) Overview of the indoor rearing facility with colony-level units; (B) Individual rearing box with a plastic mesh enclosure and syringe used for feeding; (C) Interior view of a rearing box showing a hornet nest attached to the container wall).

In total, 10 rearing rooms and approximately 10,000 colonies were surveyed, including 5300 colonies classified as healthy and 4700 colonies classified as diseased. Colonies were diagnosed as diseased when larvae showed yellowing, reduced vitality, and/or larvae

dropping from combs. To capture infection status at the colony level, both the number of diseased larvae and the number of diseased colonies were recorded. From these colonies, samples were subsequently selected for downstream experiments, with six colonies used for the oral pathogenicity assay (see Section 2.3) and twelve colonies used for gut microbiota analysis (see Section 2.4).

Sample collection involved the removal of entire nests from indoor-reared colonies, which were then transported to the laboratory for subsequent processing as well as continuous rearing and observation. Comb sections containing larvae were placed in ventilated containers with moisture control and were transported at approximately 25–30 °C. All samples reached the laboratory within 2 h and were processed immediately upon arrival.

During experimental husbandry, the original comb structure was retained, and colony-level experimental units were established. Each unit comprised one queen, approximately ten workers, and approximately twenty larvae. Rearing practice and our field observations indicated that colonies at this developmental stage are particularly susceptible to disease outbreaks. Consequently, approximately twenty larvae per colony were selected as the experimental sample size, which enabled adequate sampling at multiple time points (e.g., days 3, 5, and 7; see Sections 2.3 and 2.4) while minimizing disturbance and maintaining colony integrity and brood availability.

To minimize cross-contamination, each colony was housed in an independent enclosure. Feeding and handling tools were dedicated to each colony or disinfected between colonies. Gloves were changed between groups, and feeding was conducted beginning with control colonies and followed by infected colonies.

2.2. Sample Collection, Bacterial Isolation, and Preliminary Identification

2.2.1. Sample Collection and Processing

Larvae exhibiting typical signs, including yellowed body color, wrinkled cuticle, emaciation, and reduced vitality, were collected, and healthy larvae were sampled as controls. Specimens were placed in sterile microcentrifuge tubes, transported on ice, and delivered to the laboratory within 2 h. To ensure comparability across colonies and groups, larvae were selected from the same developmental stage (late instar) based on similar body size (approximately 1.5–2.0 cm in body length). ‘Symptomatic’ larvae were operationally defined as individuals showing visible yellowing or discoloration, reduced movement/feeding, and/or larvae detached from comb cells (dropping), as observed during routine inspection. Upon arrival, larval surfaces were disinfected with 75% ethanol for 1 min and rinsed three times with sterile distilled water. Under aseptic conditions, three diseased larvae were dissected, the gut tissues were excised, and 5 mL of sterile physiological saline was added; tissues were homogenized to prepare a gut suspension for subsequent bacterial isolation and analyses, and the suspension was stored at 4 °C for short-term preservation.

2.2.2. Isolation and Purification

The processed gut homogenate was subjected to a tenfold serial dilution to 10^{-6} , and 150 µL aliquots of the 10^{-4} , 10^{-5} , and 10^{-6} dilutions were spread onto LB agar plates. After overnight incubation at 37 °C, single colonies were picked and purified by quadrant streaking to obtain clonal isolates. For each larval gut sample, we randomly picked 8–12 colonies representing distinct colony morphotypes (based on color, size, edge, and texture) for purification. In total, 86 colonies were initially screened, and 42 isolates were obtained after repeated streaking to purity. Among these isolates, a red-pigmented *S. marcescens* strain (designated YR2) was repeatedly recovered from diseased larvae and was therefore selected for downstream phenotypic characterization and infection assays.

Colony morphology was recorded, and Gram staining was performed to determine the Gram reaction.

2.2.3. Molecular Identification (16S rRNA)

Genomic DNA was extracted from purified isolates, and the 16S rRNA gene was amplified using universal primers 27F (5'-AGAGTTTGATCCTGGCTCAG-3') and 1492R (5'-TACGGYTACCTTGTACGACTT-3') [15]. PCR conditions were as follows: 94 °C for 4 min; 35 cycles of 94 °C for 40 s, 58 °C for 40 s, and 72 °C for 90 s; a final extension at 72 °C for 10 min; hold at 16 °C. Amplicons were sequenced by Beijing Tsingke Biotechnology Co., Ltd. (Beijing, China). The resulting sequences were analyzed for homology using the NCBI online database, and a phylogenetic tree was constructed in MEGA 7.0 using the Neighbor-Joining method.

2.2.4. Scanning Electron Microscopy (SEM)

Logarithmic-phase cells were fixed in 2.5% glutaraldehyde for 12 h, dehydrated in a graded ethanol series of 30%, 50%, 70%, 90%, and 100% for 10 min at each step, replaced with isoamyl acetate, subjected to critical point drying, sputter-coated with gold, and observed and imaged using a scanning electron microscope (FEI Nova NanoSEM 450, FEI Company, Hillsboro, OR, USA).

2.3. Pathogenicity Verification of the Isolated Strain

2.3.1. Preparation of Bacterial Suspensions

A single colony of the isolated strain YR2 was inoculated into LB broth and cultured at 37 °C and 180 rpm for 18 h. To establish the relationship between optical density and viable counts, fresh LB was inoculated and incubated at 30 °C and 180 rpm to mid-log phase with an OD₆₀₀ of approximately 0.6 to 0.8, followed by tenfold serial dilution and plate counting to generate a linear regression between OD₆₀₀ and CFU per milliliter with $R^2 \geq 0.99$. The calibration was then used to set the working inoculum concentration. Based on this standard curve, an infection suspension was prepared as a 20% glucose solution containing 1.00×10^9 CFU/mL of *S. marcescens* [16].

2.3.2. Oral Infection Model and Grouping

The colony was used as the experimental unit, with each colony comprising one queen, approximately ten workers, and twenty larvae. Six colonies were established and randomly assigned to a control group (n = 3, labeled CK1 to CK3) or an infection group (n = 3, labeled GR1 to GR3). Colonies were housed in the same room but kept physically separated to prevent intermixing, and the rearing environment was maintained at 20 to 25 °C with relative humidity of 60 to 70%.

The control group received a 20% glucose solution, whereas the infection group received a 20% glucose solution containing the overnight culture of the hornet gut isolate YR2. The infection diet was prepared by mixing 10 mL of the YR2 overnight culture with 20 mL of 50% glucose solution and 20 mL of sterile water, followed by gentle homogenization. Each colony was provided 10 mL per day. After one day, both groups returned to routine feeding with pure honey, mineral water, and one locust per day. Colonies were observed for seven days, and the growth status and symptoms of queens, workers, and larvae were recorded. At each time point on day 3, day 5, and day 7, symptomatic larvae from each group were collected; if no clear symptoms were present, three larvae per colony were randomly dissected. Larvae selected for dissection were restricted to late-instar individuals of comparable size, and symptom assessment followed the operational criteria described in Section 2.2.1. Bacterial isolation and purification followed Section 2.2, and gut microbiota were profiled by 16S rRNA amplicon sequencing.

For the oral pathogenicity assay, the colony was treated as the experimental (ecological/statistical) unit. Specifically, we established two groups with three independent colonies per group: control colonies (CK1–CK3) and infected colonies (GR1–GR3). Each colony replicate contained one queen, approximately 10 workers, and 20 larvae, and larvae within the same colony were not treated as independent statistical replicates [17].

A non-infected control group was included; however, heat-killed bacteria and cell-free filtrate controls were not included in the present study and will be addressed in future work.

2.4. Gut Microbiota Sequencing and Analysis

For the microbiota experiment, the colony was treated as the experimental unit. Three groups were included with four independent colonies per group: HFYCJK1–4 (control), HFYCGRQ1–4 (mild disease), and HFYCGRZ1–4 (severe disease). For each colony, larval samples were collected at 3, 5, and 7 dpi. At each time point, symptomatic larvae were preferentially selected; if no clear symptoms were present, three late-instar larvae per colony were randomly selected for dissection. Symptom assessment followed the operational criteria described in Section 2.2.1. To evaluate the pathogenic effect of the hornet gut isolate YR2 on larvae under a fixed inoculum concentration, we modulated infection duration to establish models of differing severity. The bacterial suspension was standardized to 1.00×10^9 CFU/mL in a 20% glucose feeding solution. Daily feeding volume was held constant at 10 mL per colony and was apportioned evenly among larvae. The colony served as the experimental unit, and each colony consisted of one queen, approximately ten workers, and twenty larvae. Twelve colonies were used in total.

Based on infection status, colonies were assigned to a control group, a mild infection group, or a severe infection group. The control group received an equal volume of sterile 20% glucose solution ($n = 4$, IDs HFYCJK1 to HFYCJK4). The mild infection group received the same bacterial concentration for three consecutive days ($n = 4$, IDs HFYCGRQ1 to HFYCGRQ4). The severe infection group received the same bacterial concentration for seven consecutive days ($n = 4$, IDs HFYCGRZ1 to HFYCGRZ4). All colonies were maintained in separate enclosures to prevent mixing, with the temperature at 20 to 25 °C and relative humidity at 60 to 70%. Infected and control colonies were fed in parallel and sampled at the designated time points for downstream pathological observation and physiological measurements.

Larval mortality was monitored following oral infection. After inoculation, larvae were inspected once daily for a total period of 15 days post infection (dpi). Mortality was recorded once daily for 15 dpi. Due to the colony-level experimental design and limited independent colony replicates, we report mortality outcomes descriptively (time of first mortality, end-point mortality, and mean time to death), rather than performing Kaplan–Meier survival analysis. During the first five dpi, mortality events were recorded in detail for each sampling point; thereafter, colonies were inspected daily to follow the overall progression of mortality until 15 dpi. The time of the first observed mortality and the mean time to death (MTD) were calculated based on recorded fatality events. Individuals that remained alive at the end of the observation period were regarded as surviving larvae and were not included in MTD estimation. Activity, food intake, and mortality were recorded at fixed times each day to assess health trajectories under different infection durations. At the end of each infection phase, that is, day 3 for the mild group and day 7 for the severe group, representative diseased larvae were collected. Guts and gut contents were dissected for total DNA extraction and subsequent analyses. All samples were snap-frozen in liquid nitrogen immediately after collection and stored at -80 °C until processing.

The V3–V4 region of the 16S rRNA gene was amplified using universal primers 338F and 806R. No-template controls (NTCs) and DNA extraction blanks were included in each

PCR amplification and sequencing batch to monitor potential contamination. Purified PCR products were sequenced on an Illumina MiSeq PE300 platform. Raw reads were quality filtered with Trimmomatic, chimeras were removed with USEARCH, and downstream analyses were performed in QIIME 2. Alpha diversity (Shannon and Chao1) was calculated in QIIME 2 and is reported as mean $\pm$ SE. Group-wise comparisons were performed using the Kruskal–Wallis test with $p < 0.05$ considered significant. Beta diversity was assessed using Bray–Curtis distances and visualized by principal coordinates analysis (PCoA). Differences in community structure among groups were tested using PERMANOVA (999 permutations), and both R^2 and p values are reported. Given the limited number of independent colony-level replicates and the compositional nature of 16S amplicon data, we focused on community-level comparisons (alpha/beta diversity and PERMANOVA) and report taxonomic shifts descriptively (relative abundance of major taxa), rather than performing exploratory differential-taxon screening.

3. Results

3.1. Isolation and Identification of the Pathogenic Bacterium

To identify bacteria associated with diseased larvae, we performed culture-based isolation from larval guts. Across all samples, 86 colonies representing diverse colony morphotypes were initially screened, yielding 42 purified isolates after repeated streaking. Notably, a red-pigmented isolate consistently recovered from diseased larvae was identified as *S. marcescens* and designated YR2; because of its repeated occurrence and distinctive phenotype, YR2 was selected as the focal strain for subsequent experiments. On LB agar, YR2 formed round, entire, moist, red colonies (Figure 2A). Gram staining showed Gram-negative rods (Figure 2B), and scanning electron microscopy revealed short, smooth-surfaced rods (Figure 2C). These features are consistent with published descriptions of *S. marcescens* [18–20]. 16S rRNA gene sequencing showed greater than 99% identity to *S. marcescens*, and phylogenetic analysis placed the isolate within the *Serratia* clade (Figure 2D). Together with biochemical characteristics—oxidase negative, nitrate reduction positive, Voges–Proskauer positive, DNase positive, and gelatin liquefaction positive—the isolate was identified as *S. marcescens*.

3.2. Symptoms and Mortality of Larvae During Infection

In the oral challenge assay, outcomes are summarized at colony level ($n = 3$ colonies/group; CK1–CK3 vs. GR1–GR3); larval-level numbers are descriptive within colonies. Across the three infected colonies, larvae consistently developed yellowing and weakness, with progressive loss of normal creamy-white coloration, wrinkling of the cuticle, reduced feeding, and detachment from comb cells, and showed higher mortality than the control colonies. Such progressive decline in activity and feeding is consistent with general disease trajectories reported in insect gut–pathogen interactions [7]. Representative images of symptomatic larvae are shown in Figure 3.

Over the seven-day infection assay, colonies in the control group remained healthy, larval coloration was normal, and neither queens nor workers showed abnormalities or mortality. In the infection group, larval cuticle began to pale and activity declined from day 3 onward, accompanied by worker deaths, with 2, 4, and 2 workers dying in GR1, GR2, and GR3, respectively; by day 5, overall survival in the infection group was lower than in the controls (Figure 3A–C). *S. marcescens* was reisolated from the guts of infected larvae.

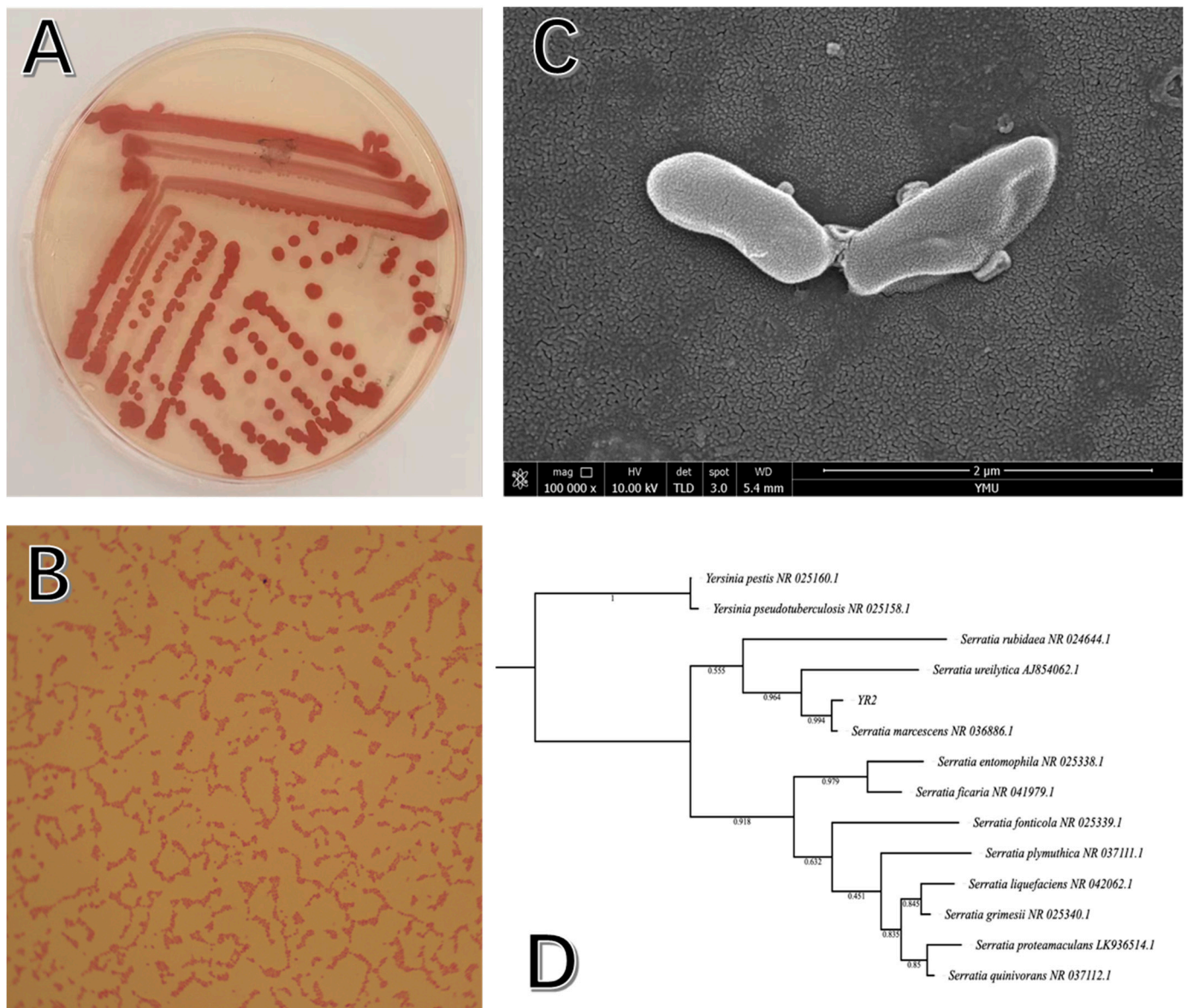


Figure 2. Isolation and identification of *S. marcescens*. (A) colony morphology observation; (B) Gram staining shows negative cells; (C) cell morphology under scanning electron microscopy (SEM); (D) phylogenetic analysis of the target isolate YR2.

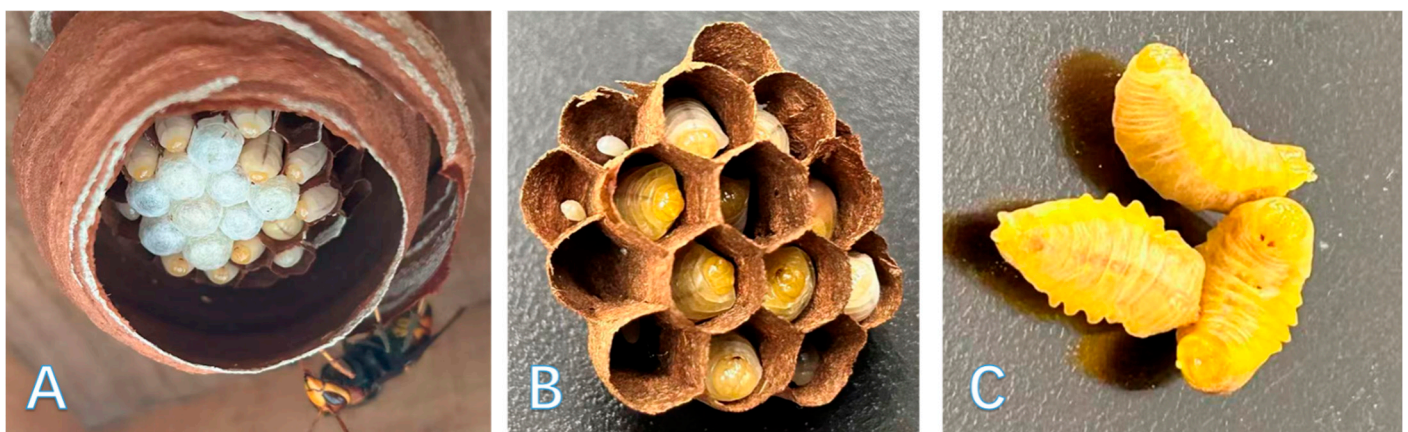


Figure 3. Comparative appearance of larvae in control and infected groups (day 5). (A) healthy hornet larvae and comb; (B) infected hornet larvae and comb; (C) infected hornet larvae.

Mortality dynamics were followed for 15 dpi after oral infection. The first larval death was observed at 4 dpi. Thereafter, mortality progressively increased, and by approximately 15 dpi nearly all larvae in the infected group had died, whereas no comparable mortality was observed in the control group. Based on the observed fatality events, the mean time to death (MTD) of infected larvae was estimated at 9.5 days. These results demonstrate that oral exposure to the isolated bacterium can reproduce disease signs and lead to substantial larval mortality under experimental conditions.

3.3. Pathogenicity of the Isolated Strain

Using larvae of *V. analis* as the host, we conducted an oral challenge with the clinically isolated *S. marcescens* strain. At day 7 post infection, larval guts were aseptically dissected and subjected to culture-based detection. *S. marcescens* was recovered from the guts of infected larvae, whereas no isolates were obtained from uninfected control larvae.

To corroborate the culture-based results, 16S rRNA amplicon sequencing was performed on the same batch of samples. No *S. marcescens* signal was detected in control larval guts (CK and GCK), while the bacterium was detected in all infected larval guts (G1–G4) (Figure 4). The sequencing results were consistent with the culture-based detection, indicating the presence of *S. marcescens* in infected larvae and its absence in controls following oral challenge.

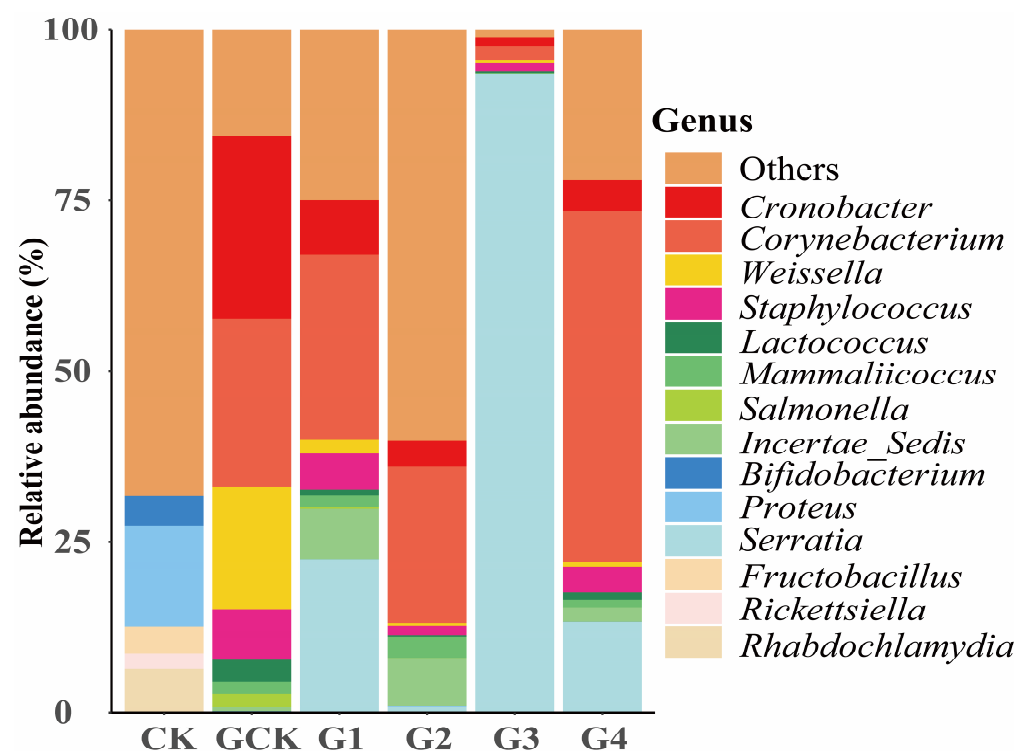


Figure 4. Distributional changes in *S. marcescens* in hornet guts before and after infection. (Relative abundance of *S. marcescens* based on 16S rRNA amplicon sequencing in control larvae (CK and GCK) and infected larvae (G1–G4). *S. marcescens* was not detected in control groups, whereas it was detected in all infected larval guts).

3.4. Correlation Between *S. marcescens* and the Gut Microbiota

To investigate the impact of *S. marcescens* infection on the gut microbiota of hornet larvae and its associations with community structure, we established three groups under identical rearing conditions: an uninfected control (HFYCJK1–HFYCJK4), a mild infection group sampled at day 3 (HFYCGRQ1–HFYCGRQ4), and a severe infection group sampled at day 7 (HFYCGRZ1–HFYCGRZ4). Larval gut samples from each group underwent 16S

rRNA amplicon sequencing. We compared the relative abundances of major taxa and related these profiles to the detection of *S. marcescens*.

Overall, the gut microbiota of control larvae was dominated by Proteobacteria (Figure 5A). As infection progressed, the relative abundance of Proteobacteria increased significantly. At broader taxonomic ranks, both Gammaproteobacteria and Enterobacteriales were elevated in the mild and severe infection groups compared with controls (Figure 5B,C). Because *S. marcescens* belongs to Enterobacteriales, changes in this order were consistent with the detection of *Serratia* in infected larvae. Alpha diversity decreased in infected groups relative to controls. Shannon index values were [Control: 3.25 ± 0.28 ; Mild: 3.25 ± 1.87 ; Severe: 2.72 ± 0.38], and Chao1 richness values were [Control: 202.86 ± 20.61 ; Mild: 301.85 ± 230.93 ; Severe: 121.94 ± 21.07] (mean $\pm$ SE). Group-wise differences were tested using the Kruskal–Wallis test (Shannon: $p = 0.309$; Chao1: $p = 0.155$), indicating non-significant reductions in within-sample diversity associated with infection.

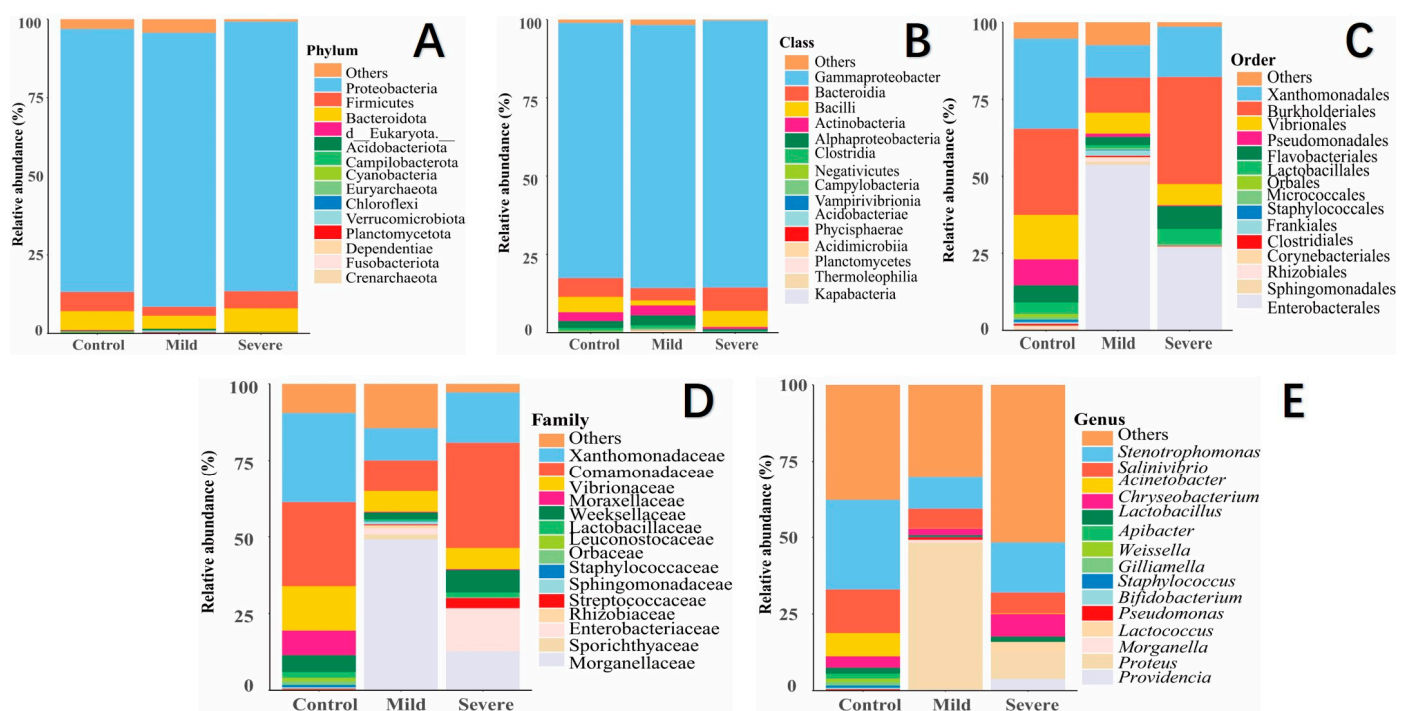


Figure 5. Taxonomic composition and relative abundance of larval hornet gut microbiota under different infection states. (A) relative abundance at the phylum level; (B) composition and relative abundance at the class level; (C) relative abundance at the order level across infection stages; (D) comparative analysis at the family level across infection stages; (E) relative abundance at the genus level across infection stages.

Consistently, α -diversity indices (Shannon and Chao1) tended to decrease in infected larvae relative to controls, indicating reduced within-sample diversity.

At finer taxonomic levels, Morganeliaceae and the genus *Proteus* increased in the infected groups (Figure 5D,E). These shifts suggest responsive fluctuations among ecologically proximate members that may compete or interact with *S. marcescens*. Integrating culture-based reisolation with amplicon data, we detected *S. marcescens* in infected larvae alongside a community shift toward Proteobacteria and their representative lineages from phylum to genus levels, whereas the control group showed no detection of *S. marcescens* and maintained a stable community composition. In addition, β -diversity based on Bray–Curtis distances showed significant differences in overall community structure, particularly between the control and severe-infection groups (PERMANOVA: $R^2 = 0.212$, $p = 0.034$),

whereas differences between control and mild infection were weaker ($R^2 = 0.361$, $p = 0.149$). ANOSIM analyses yielded consistent tendencies.

At finer taxonomic levels, Morganellaceae and the genus *Proteus* increased in the infected groups (Figure 5D,E), suggesting responsive fluctuations among ecologically proximate members during infection-associated community shifts. In addition, beta diversity based on Bray–Curtis distances revealed differences in overall community structure among groups (PERMANOVA, 999 permutations). Specifically, the control and severe-infection groups differed significantly ($R^2 = 0.212$, $p = 0.034$), whereas the control and mild-infection groups showed a weaker, non-significant separation ($R^2 = 0.361$, $p = 0.149$).

Taken together, the detection/expansion of *S. marcescens* in infected larvae was associated with a dysbiosis pattern characterized by increased dominance of Proteobacteria (particularly Gammaproteobacteria and Enterobacterales) and concurrent increases in Morganellaceae and *Proteus* (Figure 5A–E). These taxonomic shifts, together with alpha diversity metrics and PERMANOVA results, provide quantitative support for infection-associated dysbiosis under our challenge conditions.

4. Discussion

Together, our isolation, oral challenge, and reisolation results provide converging evidence that *S. marcescens* is associated with increased larval mortality and consistent disease signs in indoor-reared *V. analis*, supporting it as a strong candidate pathogen in this system, although further strain-level validation is required to establish causality. Infected larvae developed characteristic phenotypes (e.g., yellowing or discoloration, reduced activity and feeding, and elevated mortality), and a phenotypically consistent *S. marcescens* isolate was recovered from larval guts following challenge, whereas *Serratia* was not detected (or was rarely detected) in controls. These observations align with evidence from other Hymenoptera; notably, *S. marcescens* strains display measurable pathogenicity in honey bees, with outcomes that can depend on strain identity and infection context [11,21]. In addition, work in bumblebees indicates that colonization by gut-associated bacterial pathogens is shaped by core symbionts, inoculation timing, and diet, reinforcing the idea that disease emergence can be strongly context-dependent [22]. Taken together, our findings extend the relevance of *S. marcescens* to hornet larval health and highlight its potential to act as a disease-associated pathobiont in managed colonies.

The homeostasis of the gut microbiota is critical to insect health, supporting nutrient assimilation, immune regulation, and defense against exogenous microbes [7,23]. Consistent with this framework, insect gut immunity and barrier function are increasingly recognized as key determinants of whether opportunistic bacteria remain controlled or expand during stress and infection [22]. In this study, healthy *V. analis* larvae harbored gut communities dominated by Proteobacteria, with Enterobacterales and Morganellaceae as prominent lineages. Following challenge, community structure shifted and a dysbiosis pattern emerged, characterized by increased dominance of Enterobacterales and detectable expansion of *Serratia* [24]. From an ecological perspective, blooms of related bacterial groups in the gut can occur simultaneously when the gut environment is perturbed, reflecting shared metabolic or ecological advantages under altered conditions rather than a single-cause endpoint [25,26]. A plausible interpretation is that challenge perturbs the gut competitive landscape and resource availability, enabling rapid niche takeover by fast-growing facultative anaerobes; in parallel, *S. marcescens* may further amplify community drift via resource competition and secreted activities (e.g., exoenzymes), potentially interacting with host defenses and barrier integrity [22,25]. Importantly, these microbiota changes should be interpreted as disease-associated dysbiosis correlated with disease progression, rather than definitive proof that community shifts alone cause pathology.

These results have practical implications for indoor hornet husbandry and outbreak management. Compared with natural settings, indoor rearing can intensify stressors (handling, density, and dietary shifts) and increase transmission opportunities through shared feeding routes, which may facilitate opportunists transitioning from low-level colonizers to outbreak-associated taxa [21,27]. Accordingly, routine monitoring strategies that combine phenotypic screening with microbiota-based warning signals, such as increased Enterobacteriales dominance and detectable *Serratia*, may help identify risk states before large-scale larval loss occurs. Our results also reinforce the importance of strict separation among colony units, dedicated or disinfected tools, and standardized handling order to reduce cross-contamination.

Several limitations should be acknowledged. First, strain identity between the inoculum and reisolates was not confirmed at the genome level in the present study; future work will address this using MLST and/or whole-genome comparisons and incorporate dose-response designs. Second, we did not include heat-inactivated bacteria or sterile cell-free filtrates; therefore, we cannot exclude the possibility that preformed or secreted factors contributed to mortality in addition to active infection. Future studies will incorporate heat-killed and filtrate controls, longer time-series sampling, and quantitative measures of clinical severity to better disentangle live infection versus secreted-factor effects and to link dysbiosis dynamics with pathology. Importantly, the conclusions of the current study are limited to association and experimental support under our challenge conditions, rather than definitive attribution of mortality to active infection alone.

5. Conclusions

Focusing on larvae of the hornet *V. analis*, we isolated and identified *S. marcescens* from the guts of naturally diseased individuals and characterized the isolate by colony and cellular morphology, 16S rRNA sequencing, and scanning electron microscopy; in an oral infection model, we reproduced characteristic phenotypes, including larval yellowing, reduced vigor, and mortality, and we reisolated *S. marcescens* phenotypically consistent to the inoculum from infected guts. At the community level, infection was accompanied by a decrease in α diversity and a dysbiotic shift characterized by enrichment of Proteobacteria, indicating that colonization by *S. marcescens* is closely associated with disruption of host gut homeostasis. Overall, our results support *S. marcescens* as a strong candidate pathogen associated with larval disease and mortality in indoor-reared *V. analis*, while additional controls (e.g., heat-killed bacteria and cell-free filtrates) are needed to disentangle active infection from preformed/secreted factor effects.

This study provides empirical support for pathogen surveillance and microbiota-informed health management in indoor hornet husbandry. Future work will strengthen causal inference by incorporating strain-level genotyping (MLST/WGS), heat-killed and cell-free filtrate controls, and standardized dose-response and time-course designs to clarify the relative contributions of active infection and secreted factors and to better resolve microbiota dynamics during disease progression.

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Data Availability Statement: The 16S rRNA amplicon sequencing data generated in this study have been deposited in the NCBI Sequence Read Archive (SRA) under accession number PX602940. All other data supporting the conclusions of this article are included in the article.

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